

TCLK Event Definitions 31-Aug-2021

[Click here for definitions from 19-June-2012](#) (before the decommissioning of Tev events)

Event	Source	System	Delay name	Description	HCRM channel
\$00	TLG	CLK		Super Cycle and Master Clock Reset";marks start of supercycle	TCLK \$06 N5, HCRM #16 CH0
\$01					
\$02	C002	CLK		Time Plot Timestamp Reset:generated once every 50,000,000 TCLK cycles (every 5 seconds); not synchronized to any other event	TCLK \$06 N19, HCRM #2 CH1
\$03	BSSB	MTA		MTA Beam Permitted	TCLK \$06 N19, HCRM #2 CH4
\$04	TLG	CLK		MTA Pulse on next \$11	TCLK \$06 N8, HCRM #13 CH9
\$05	C377	CLK	T:UFEVT	User Friendly Event; intended for short term use; user selectable delay and reference	TCLK \$06 N14, HCRM #7 CH13
\$06				(reserved for 60 Hz synchronized to AC line)	
\$07	MCO	CLK		720 Hz; poorly synchronized to AC line	TCLK \$06 N20, HCRM #1 CH4
\$08					
\$09	BSSB	NTF		Permitted NTF Beam Cycle; coincident with \$11	TCLK \$06 N9, HCRM #12 CH8
\$0A	BSSB	LINAC		Permitted 400 MeV Linac Studies Cycle; coincident with \$11	TCLK \$06 N14, HCRM #7 CH1
\$0B	C377	BSTR	B:DATRDRY	Booster Beam Sample Data Ready; referenced to \$10	TCLK \$06 N16, HCRM #5 CH15
\$0C	TLG	CLK		15 Hz; synchronized to GMPS BMIN Delayed	TCLK \$06 N7, HCRM #14 CH12
\$0D	C377	BSTR	T:TD0Dn n=1...8	Booster Beam Sample Time; "or" of 8 377 channels referenced to \$13,\$14,\$15,\$16,\$17,\$19,\$1C,\$1D	TCLK \$06 N9, HCRM #12 CH12
\$0E	TLG	BSTR		First Booster Beam Reset After a Pre-pulse Cycle	TCLK \$06 N5, HCRM #16 CH7
\$0F	MCO	CLK		15 Hz; synchronized to A phase of AC line in MAC room	TCLK \$06 N20, HCRM #1 CH5
\$10	hrdw	BSTR		Generic Booster Reset for Accelerating Beam Cycle; "or" of \$13,\$14,\$15,\$16,\$17,\$19,\$1C,\$1D	TCLK \$06 N18, HCRM #3 CH12
\$11	TLG	BSTR		Booster Reset for Null Cycle; generated at GMPS BMIN Delayed	TCLK \$06 N7, HCRM #14 CH1

\$12	TLG	BSTR		Booster Reset for Beam Pre-pulse Cycle; generated at GMPS BMIN Delayed; always preceeded by \$08	TCLK \$06 N7, HCRM #14 CH2
\$13	TLG	BSTR		Booster Reset for Fixed Target Beam Cycle; generated at GMPS BMIN Delayed	TCLK \$06 N7, HCRM #14 CH3
\$14	TLG	BSTR		Booster Reset for MI/Muon Studies; generated at GMPS BMIN Delayed	TCLK \$06 N7, HCRM #14 CH4
\$15	TLG	BSTR		Booster Reset for NUMI Beam Cycle; generated at GMPS BMIN Delayed	TCLK \$06 N7, HCRM #14 CH5
\$16	TLG	BSTR		Booster Reset for MI Studies; generated at GMPS BMIN Delayed	TCLK \$06 N7, HCRM #14 CH6
\$17	TLG	BSTR		Booster Reset for Booster Studies Cycle; generated at GMPS BMIN Delayed	TCLK \$06 N7, HCRM #14 CH7
\$18	C377	BSTR	B:SAMPLT	Booster Sample Time; used for Booster data sampling in front-end; referenced to \$10,\$11,\$12	TCLK \$06 N15, HCRM #6 CH10
\$19	TLG	BSTR		Booster Reset for NUMI Beam Cycle; generated at GMPS BMIN Delayed	TCLK \$06 N7, HCRM #14 CH8
\$1A	TLG	NTF		Parasitic Beam Permit; made @ positive transition of TLG output	TCLK \$06 N5, HCRM #16 CH12
\$1B	TLG	NTF		Parasitic Beam Inhibit; made @ negative transition of TLG output	TCLK \$06 N7, HCRM #14 CH15
\$1C	TLG	BSTR		Booster Reset for MI/RR Studies; generated at GMPS BMIN Delayed	TCLK \$06 N7, HCRM #14 CH9
\$1D	TLG	BSTR		Booster Reset for MiniBooNe Beam Cycle; generated at GMPS BMIN Delayed	TCLK \$06 N7, HCRM#14 CH10
\$1E	C064	BSTR		Booster Extraction Sync (BES) for a Pre-pulse Cycle"; generated by BES during a Booster \$12 pre-pulse cycle	TCLK \$06 N16, HCRM#5 CH5
\$1F	C064	BSTR		Booster Extraction Sync (BES) for a Beam Cycle";generated by a BES during a Booster beam cycle	TCLK \$06 N16, HCRM#5 CH6
\$20	TLG	MI		MI Reset for 120 GeV Short Flatop Fixed Target Beam	TCLK \$06 N6, HCRM #15 CH0
\$21	TLG	MI		MI Reset for 120 GeV Long Flatop Fixed Target Beam	TCLK \$06 N6, HCRM #15 CH1
\$22	C377	MI	I:RPSTx x=20,...	Start of MI Ramp Up; "or" of 8 377 channels with separate references to \$20,\$21,\$23,\$29,\$2A,\$2B,\$2D,\$2E	TCLK \$06 N18, HCRM #3 CH11
\$23	TLG	MI		MI Reset for NUMI Beam Cycle	TCLK \$06 N6, HCRM #15 CH2
\$24	sftw (I67)	MI		MI BP/Abort System Reset	TCLK \$06 N11, HCRM #10 CH6
\$25	C377	MI	I:FTSTx x=20,...	Start of MI Ramp Flatop; "or" of 8 377 channels with separate references to \$20,\$21,\$23,\$29,\$2A,\$2B,\$2D,\$2E	TCLK \$06 N18, HCRM #3 CH14

\$26	C377	MI	I:EOBx x=20,...	End of MI Ramp Flatop; "or" of 8 377 channels with separate references to \$20,\$21,\$23,\$29,\$2A,\$2B,\$2D,\$2E	TCLK \$06 N18, HCRM #3 CH15
\$27	C004	MI		MI Beam Aborted;requested on fall of Abort/Permit loop	TCLK \$06 N20, HCRM #1 CH10
\$28	C064	MI		MI 8GeV Injected Beam from BSTR; generated by \$1F on \$13, \$14, \$15, \$16, \$19 or \$1C cycles	TCLK \$06 N12, HCRM #9 CH5
\$29	TLG	MI		MI Reset for MI Studies/120 GeV to Muon Cycle	TCLK \$06 N6, HCRM #15 CH9
\$2A	TLG	MI		MI Reset for NuMI Beam with RR Cycle	TCLK \$06 N6, HCRM #15 CH10
\$2B	TLG	MI		MI Reset for MI Studies	TCLK \$06 N6, HCRM #15 CH11
\$2C	C377	MI	T:TD2Cn n=0...7	MI Sample Time; "or" of 8 377 channels with separate references to \$20,\$21,\$23,\$29,\$2A,\$2B,\$2D,\$2E	TCLK \$06 N17, HCRM #4 CH15
\$2D	TLG	MI		MI Reset for 8 GeV Beam to Muon/MI Studies	TCLK \$06 N6, HCRM #15 CH13
\$2E	TLG	MI		MI Reset for MI Studies	TCLK \$06 N6, HCRM #15 CH14
\$2F	C377	MI	I:ACUPx x=20,...	MI Cleanup; "or" of 8 377 channels with separate references to \$20,\$21,\$23, \$29,\$2A,\$2B,\$2D,\$2E	TCLK \$06 N20, HCRM #1 CH11
\$30	TLG	SWYD		Switchyard Reset for 120 GeV, Long Spill Extracted Beam	TCLK \$06 N8, HCRM #13 CH3
\$31	C377	SWYD	S:RESET	Switchyard Reset for Extracted Beam Cycle; \$30 & \$32 are usual references	TCLK \$06 N15, HCRM #6 CH8
\$32	TLG	SWYD		Switchyard Pre Pulse	TCLK \$06 N8, HCRM #13 CH8
\$33	C377	SWYD	T:TD33x x=0..7	Switchyard Fast Spill/Ping	TCLK \$06 N13, HCRM #8 CH1
\$34	C070	SWYD	S:SWIC34	Switchyard SWIC Sample Time; multiple events timed from an "initialize" reference; "initialize" is sourced by S:TSWIC timing channel; 070 is at TCLK \$04 N14-15; clock is 100KHz; companion 068 is at N13	TCLK \$06 N16, HCRM #5 CH7
\$35	C377	SWYD	S:BST	Switchyard Beam Sample Time; \$31 is normal reference	TCLK \$06 N14, HCRM #7 CH14
\$36	C377	SWYD	S:QXREND	Switchyard Post-Beam Sample Time; \$31 is normal reference	TCLK \$06 N14, HCRM #7 CH15
\$37	C377	SWYD	S:INTEND	Switchyard Read Intensity Monitors	TCLK \$06 N14, HCRM #7 CH12
\$38	sftw (S67)	SWYD		Switchyard Beam Permit System Reset	TCLK \$06 N12, HCRM #9 CH13
\$39	MIBS	SWYD/MI		TCLK reflection of MIBS \$75: 120GeV Proton extraction from MI to SYWD	TCLK \$06 N12, HCRM #9 CH14

\$3A	C377	SWYD	T:TD3An n=1...8	Switchyard BPM Write Profile Memory; "or" of 8 377 channels with common \$31 reference	TCLK \$06 N11, HCRM #10 CH7
\$3B	C377	SWYD	T:TD3B	Switchyard BPM Write Display Frame	TCLK \$06 N11, HCRM #10 CH13
\$3C	C377	SWYD	S:VAREVT	Switchyard Variable Event	TCLK \$06 N17, HCRM #4 CH6
\$3D	C377	SWYD	S:ANAEVT	Switchyard Analog Alarm Event	TCLK \$06 N18, HCRM #3 CH4
\$3E	C004	SWYD		SWYD Permit has Dropped; requested on fall of permit	TCLK \$06 N18, HCRM #3 CH3
\$3F	C377	SWYD	T:EOISWD	End of Switchyard beam injection	TCLK \$06 N13, HCRM #8 CH2
\$40				Decommissioned	
\$41				Decommissioned	
\$42				Decommissioned	
\$43				Decommissioned	
\$44				Decommissioned	
\$45				Decommissioned	
\$46				Decommissioned	
\$47				Decommissioned	
\$48				Decommissioned	
\$49				Decommissioned	
\$4A				Decommissioned	
\$4B				Decommissioned	
\$4C	BSTR	C377	B:VAREVT	Booster Variable Event	TCLK \$06 N12, HCRM #9 CH0
\$4D				Decommissioned	
\$4E				Decommissioned	
\$4F				Decommissioned	
\$50	C377	MI	G:SMRSTD	Begin MI Studies	TCLK \$06 N16, HCRM #5 CH11
\$51	C377	MI	G:EMRSTD	End MI Studies	TCLK \$06 N16, HCRM #5 CH12
\$52	BSSB	MISC		HEP Beam Permitted by BSSB;generated at \$10 when beam to Booster is permitted	TCLK \$06 N12, HCRM #9 CH11
\$53	BSSB	MISC		HEP Beam Inhibited by BSSB;generated at \$10 when beam to Booster is inhibited	TCLK \$06 N12, HCRM #9 CH12
\$54				Decommissioned	
\$55				Decommissioned	
\$56					
\$57	MIBS	MI		TCLK Reflection of MIBS \$77: Proton transfer from Booster to Main Injector	TCLK \$06 N13, HCRM #8 CH0

\$58	C377	RR	T:TD58	RR Gap Clearing Kicker C479 Trigger Clear	TCLK \$06 N16, HCRM #5 CH0
\$59				Decommissioned	
\$5A					
\$5B				Decommissioned	
\$5C				Decommissioned	
\$5D				Decommissioned	
\$5E				Decommissioned	
\$5F				Decommissioned	
\$60				Decommissioned	
\$61				Decommissioned	
\$62				Decommissioned	
\$63				Decommissioned	
\$64				Decommissioned	
\$65				Decommissioned	
\$66				Decommissioned	
\$67	C004	P1		P1 Permit has Dropped; requested on fall of permit	TCLK \$06 N18, HCRM #3 CH0
\$68				Decommissioned	
\$69				Decommissioned	
\$6A				Decommissioned	
\$6B				Decommissioned	
\$6C				Decommissioned	
\$6D				Decommissioned	
\$6E				Decommissioned	
\$6F				Decommissioned	
\$70	C064	8GEV	T:TD70S T:TD70R	First \$1D of chain	TCLK \$06 N12, HCRM #9 CH3
\$71				Decommissioned	
\$72	C377	BSTR	B:BPMEOB	Booster BPM - End of Beam	TCLK \$06 N11, HCRM #10 CH9
\$73				Decommissioned	
\$74				Decommissioned	
\$75				Decommissioned	
\$76				Decommissioned	
\$77	C004	P2		P2 Permit has Dropped; requested on fall of permit	TCLK \$06 N18, HCRM #3 CH1
\$78				Decommissioned	
\$79	C064/C377	MI	I:BPMPFB	MI BPM - Prepare for Beam; gated by MI Beam Permit	TCLK \$06 N15, HCRM #6 CH7
\$7A	C070	MI	I:BPMWPM	MI BPM - Write Profile Memory; multiple events timed from an "initialize" reference which is the "or" of	TCLK \$06 N16, HCRM #5 CH3

				\$20,\$21,\$23,\$29,\$2A,\$2B,\$2D,\$2E; 070 is at TCLK \$07 N7-8; 070 clock is 100KHz, companion 068 is at N6	
\$7B	C377	MI	I:BPDFx x=20,...	MI BPM - Write Display Frame; "or" of 8 377 channels with separate references to \$20,\$21,\$23,\$29,\$2A,\$2B,\$2D,\$2E	TCLK \$06 N18, HCRM #3 CH5
\$7C	C279s	MI	I:BPFTx	MI BPM Flash Trigger; "or" of 12 279 ch., 5 with separate MIBS refs. (75,79,7A,7D+7E,D8) 4 with separate RRBS refs. (A0,A2,A3,A7) 1 referenced to BES, 2 referenced to 377 channels I:BPFTT1 & I:BPFTT2; BES ref gated off during \$12, \$17 or \$1D	TCLK \$06 N15, HCRM #6 CH0
\$7D	C377	MI	I:BPMBUN	MI BPM - Set Bunch Mode	TCLK \$06 N13, HCRM #8 CH13
\$7E	C377	BSTR	B:BPMBCO	Booster BPM - Begin Closed Orbit	TCLK \$06 N11, HCRM #10 CH10
\$7F	C377	BSTR	B:BPMFTn n=1...8	Booster BPM - Flash Trigger; "or" of 8 377 channels	TCLK \$06 N12, HCRM #9 CH1
\$80	TLG	MUON		Muon Reset for non 8GeV protons to Muon via MI	TCLK \$06 N5, HCRM #16 CH15
\$81	MIBS	MUON/MI		TCLK reflection of MIBS \$79: Non 8GeV transfer from MI to Muon	TCLK \$06 N17, HCRM #4 CH5
\$82	C377	MUON	T:TD82	Muon Toroid Sample Trigger	TCLK \$06 N16, HCRM #5 CH14
\$83	C377	MUON	T:TD83	Start DR Resonant Ext	TCLK \$06 N17, HCRM #4 CH9
\$84	C377	MUON	D:DRPFB	Muon Delivery Ring prepare for beam	TCLK \$06 N14, HCRM #7 CH10
\$85	TLG	MUON		Muon Reset for 8 GeV protons to Muon via MI	TCLK \$06 N5, HCRM #16 CH6
\$86	C377	MUON	D:EOBx x=80,85,93,94	Muon end of Beam Operations	TCLK \$06 N19, HCRM #02 CH8
\$87	C004	MUON		Muon Beam Permit has Dropped	TCLK \$06 N20, HCRM #01 CH7
\$88	sftw (P67)	MUON		Muon Beam Permit System Reset	TCLK \$06 N14, HCRM #7 CH0
\$89	C377	MUON	D:ACUPx x=80,85,93,94	Muon Abort Cleanup	TCLK \$06 N19, HCRM #02 CH9
\$8A					
\$8B				Decommissioned	
\$8C	C377	MUON/MI	T:TD8Cn n=0...5	Muon BPM/BLM Flash Trigger; "or" of 6 377 channels	TCLK \$06 N15, HCRM #6 CH2
\$8D	C377	MUON	D:VAREVT	Muon variable event 1	TCLK \$06 N16, HCRM #5 CH13
\$8E	TLG	MUON/MI		P1 & P2 Beamline Reset	TCLK \$06 N8, HCRM #13 CH6

\$8F	GPS	CLK		1 Hz; Generated by GPS RCVR in computer room	TCLK \$06 N20, HCRM #1 CH15
\$90	TLG	MUON		Muon Reset #1 for 8 GeV Protons from RR to M3 Line	TCLK \$06 N5, HCRM #16 CH4
\$91	TLG	MUON		Muon Reset #2 for 8 GeV Protons from RR to M3 Line	TCLK \$06 N5, HCRM #16 CH5
\$92	C377	MUON	T:TD92	Muon P1/P2 Ramp Reset	TCLK \$06 N12, HCRM #9 CH2
\$93	TLG	MUON		"Muon Reset #1 for 8 GeV Protons from RR to Muon Target"	TCLK \$06 N5, HCRM #16 CH8
\$94	TLG	MUON		"Muon Reset #2 for 8 GeV Protons from RR to Muon Target"	TCLK \$06 N5, HCRM #16 CH9
\$95	C377	MUON	T:TD95	DR IKIK Charge Start	TCLK \$06 N14, HCRM #7 CH9
\$96	C377	MUON	T:TD96	RR->Muon Transfer #6	TCLK \$06 N12, HCRM #9 CH8
\$97	C377	MUON	T:TD97	RR->Muon Transfer #7	TCLK \$06 N12, HCRM #9 CH9
\$98	C377	MUON	T:TD98	RR->Muon Transfer #8	TCLK \$06 N12, HCRM #9 CH10
\$99	MIBS	MUON/MI		TCLK Reflection of MIBS \$7E: MI to Muon 8 GeV Transfer	TCLK \$06 N13, HCRM #8 CH8
\$9A	C377	MUON	T:TD9A	RR->Muon Transfer #1	TCLK \$06 N13, HCRM #8 CH9
\$9B	C377	MUON	T:TD9B	RR->Muon Transfer #2	TCLK \$06 N13, HCRM #8 CH10
\$9C	C377	MUON	T:TD9C	RR->Muon Transfer #3	TCLK \$06 N13, HCRM #8 CH11
\$9D	C377	MUON	T:TD9D	RR->Muon Transfer #4	TCLK \$06 N13, HCRM #8 CH12
\$9E	C377	MUON	T:TD9E	RR->Muon Transfer #5	TCLK \$06 N12, HCRM #9 CH7
\$9F	C377	MUON	T:TD9F	RR->Muon Kicker Trigger Clear	TCLK \$06 N13, HCRM #8 CH15
\$A0	C377	MI	I:BLMPFB	MI BLM - Prepare for Beam	TCLK \$06 N15, HCRM #6 CH4
\$A1	C377	RR	R:BLMPFB	RR BLM - Prepare for Beam	TCLK \$06 N15, HCRM #6 CH5
\$A2	C377	EXAR	G:FTARM1	Fixed Target Sample Trigger	TCLK \$06 N15, HCRM #6 CH11
\$A3	C377	EXAR	T:TDA3	Fixed Target Sample Trigger #2	TCLK \$06 N17, HCRM #4 CH7
\$A4	C377	NUMI	T:NMICST	NuMI Cycle Sample Trigger; \$A5 reference	TCLK \$06 N17, HCRM #4 CH8
\$A5	TLG	NUMI		NUMI Reset for Beam	TCLK \$06 N8, HCRM #13 CH1

\$A6	C004	NUMI		NUMI Beam Permit has Dropped; requested on fall of permit loop	TCLK \$06 N19, HCRM #2 CH3
\$A7	sftw	EXAR		No Beam to Experimental Areas; consult W Kissel and B Hendricks	TCLK \$06 N10, HCRM #11 CH1
\$A8	sftw (E39)	NUMI		NUMI Beam Permit /Abort System Reset	TCLK \$06 N10, HCRM #11 CH14
\$A9	MIBS	NUMI/MI		TCLK reflection of MIBS \$74: 120Gev Proton Extraction from MI to NUMI	TCLK \$06 N11, HCRM #10 CH15
\$AA					
\$AB					
\$AC	TLG	CLK		MI Abort Charge Event	TCLK \$06 N7, HCRM #14 CH14
\$AD	C377	NUMI	T:TDAD	NUMI Reset for Beamline Ramp & Extracted Beam	TCLK \$06 N19, HCRM #2 CH13
\$AE	TLG	NUMI		NUMI Beamline Ramp Rest	TCLK \$06 N8, HCRM #13 CH5
\$AF	C377	NUMI	T:EOINMI	End of NUMI beam injection	TCLK \$06 N13, HCRM #8 CH3
\$B0	C064	MB		BNB Beam On Target	TCLK \$06 N10, HCRM #11 CH3
\$B1	C064	MB		BNB Beam Off Target	TCLK \$06 N10, HCRM #11 CH4
\$B2	C064/C377	RR	R:BPMPFB	RR BPM - Prepare for Beam	TCLK \$06 N15, HCRM #6 CH13
\$B3	C070	RR	R:BPMWPM	RR BPM - Write Profile Memory; multiple events timed from an "initialize" reference which is the "or" of \$E0,\$E1,\$E1,\$E3,\$E9; 070 is at TCLK\$07 N4,5; 070 clock is 100KHz, companion 068 is at N3	TCLK \$06 N16, HCRM #5 CH4
\$B4	C377	RR	R:BPDPnn	RRBPM - Write Display Frame; "or" of 8 377 channels with separate references to \$E0, \$E1, \$E2, \$E3, \$E9	TCLK \$06 N18, HCRM #3 CH2
\$B5	C279s	RR	R:BPFTx	RR BPM Flash Trigger; "or" of 5 279 channels, 2 with separate RRBS refs. (A3,A7) 1 referenced to BES, 2 referenced to 377 channels R:BPFTT1 & R:BPFTT2; BES ref gated off during \$12, \$17 or \$1D	TCLK \$06 N15, HCRM #6 CH14
\$B6	C377	RR	R:BPMBUN	RR BPM - Set Bunch Mode	TCLK \$06 N13, HCRM #8 CH14
\$B7		MB		MiniBooNE Beam Permit has dropped	Not Implemented
\$B8	sftw (E29)	MB		MiniBooNe Beam Permit /Abort System Reset	TCLK \$06 N9, HCRM #12 CH11
\$B9	sftw	MI/RR		MI/RR Periodic Data Trigger	TCLK \$06 N9, HCRM #12 CH7
\$BA	MCO	CLK		20Hz; synchronized to A phase of AC line in MAC room	TCLK \$06 N20, HCRM #1 CH14

\$BB	sftw (D103)	CLK		TCLK diagnostics - Start	TCLK \$06 N9, HCRM #12 CH4
\$BC	sftw (D103)	CLK		TCLK diagnostics - Stop	TCLK \$06 N9, HCRM #12 CH5
\$BD	C377	MI8	I:M8SMPL	MI-8 Line BLM Sample Time	TCLK \$06 N9, HCRM #12 CH13
\$BE	TLG	MI		Booster beam to Recycler	TCLK \$06 N8, HCRM #13 CH11
\$BF	TLG	MI		Booster Beam to Main Injector/MiniBooNE	TCLK \$06 N8, HCRM #13 CH12
\$C0				Decommissioned	
\$C1				Decommissioned	
\$C2				Decommissioned	
\$C3				Decommissioned	
\$C4				Decommissioned	
\$C5				Decommissioned	
\$C6				Decommissioned	
\$C7				Decommissioned	
\$C8					
\$C9				Decommissioned	
\$CA				Decommissioned	
\$CB				Decommissioned	
\$CC				Decommissioned	
\$CD				Decommissioned	
\$CE				Decommissioned	
\$CF				Decommissioned	
\$D0				Decommissioned	
\$D1				Decommissioned	
\$D2				Decommissioned	
\$D3					
\$D4				Decommissioned	
\$D5				Decommissioned	
\$D6				Decommissioned	
\$D7				Decommissioned	
\$D8					
\$D9				Decommissioned	
\$DA	C064	RR	T:TDDAS T:TDDAR	RBEX Trigger	TCLK \$06 N15, HCRM #6 CH3
\$DB				Decommissioned	
\$DC				Decommissioned	
\$DD	C064	BSTR	T:TDDDS T:TDDDR	8 GeV BPM Sample Time	TCLK \$06 N15, HCRM #6 CH1

\$DE	TLG	RR		RR Reset for beam to MI	TCLK \$06 N6, HCRM #15 CH15
\$DF					
\$E0	TLG	RR		RR Reset for RR Studies	TCLK \$06 N6, HCRM #15 CH4
\$E1	TLG	RR		RR Reset for MI Studies	TCLK \$06 N6, HCRM #15 CH5
\$E2	TLG	RR		RR Reset for Fixed Target	TCLK \$06 N6, HCRM #15 CH6
\$E3	TLG	RR		RR Reset for NuMI	TCLK \$06 N6, HCRM #15 CH7
\$E4	C064	RR		RR 8 GeV Injected Beam from BSTR	TCLK \$06 N12, HCRM #9 CH6
\$E5				Decommissioned	
\$E6	C377	RR	R:EOBnn	RR End of Beam	TCLK \$06 N18, HCRM #3 CH7
\$E7	C004	RR		RR Beam Aborted; requested on fall of Abort/Permit Loop	TCLK \$06 N20, HCRM #1 CH13
\$E8	sftw (R67)	RR		RR BP/Abort System Reset	TCLK \$06 N17, HCRM #4 CH11
\$E9	TLG	RR		RR Reset for Muon	TCLK \$06 N6, HCRM #15 CH8
\$EA	TLG	MTA		MTA Pre-Event	TCLK \$06 N8, HCRM #13 CH10
\$EB				Decommissioned	
\$EC				Decommissioned	
\$ED	C377	RR	R:VAREVT	RR Variable Event	TCLK \$06 N17, HCRM #4 CH12
\$EE				Decommissioned	
\$EF	C377	RR	R:ACUPnn	RR Abort Cleanup	TCLK \$06 N20, HCRM #1 CH12
\$F0				Decommissioned	
\$F1	TLG	BSTR		Clear \$0E Holdoff	TCLK \$06 N5, HCRM #16 CH7
\$F2				Decommissioned	
\$F3	RRBS	MI/RR		TCLK reflection of RRBS \$A3: extract protons from RR into MI	TCLK \$06 N10, HCRM #11 CH9
\$F4					
\$F5				Decommissioned	
\$F6	RRBS	MUON/RR		TCLK reflection of RRBS \$A6: 8 GeV proton transfer from RR to Muon	TCLK \$06 N11, HCRM #10 CH5
\$F7	RRBS	RR		TCLK reflection of RRBS \$A7: Proton transfer from Booster to Recycler	TCLK \$06 N9, HCRM #12 CH9
\$F8					
\$F9				Decommissioned	

\$FA	MIBS	MI/RR		TCLK reflection of MIBS \$ED: MIBS/RRBS event request denied	TCLK \$06 N9, HCRM #12 CH14
\$FB					
\$FC				Booster Low Level Easter Egg	
\$FD					
\$FE	xxx	CLK		No-Op -- never generated	
\$FF	xxx	CLK		No-Op -- sometimes generated	

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